

08.06 Using Logarithms to Solve Exponential Equations

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
Exponential And Logarithmic Models	Decide whether the unknown is in an exponent, inside a logarithm, or in a model parameter.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

Example 1: Identify the model

Problem

$$A(t) = 500(1.06)^t$$

How to solve it

Write the known quantities and restrictions.

$$A(t) = A_0 b^t$$

Check the result against the original statement.

Example 2: Apply the key rule

Problem

$$3^x = 20$$

How to solve it

Write the known quantities and restrictions.

$$\log_b(x) = y \iff b^y = x$$

Check the result against the original statement.

Example 3: Check the result

Problem

$$\log_b\left(\frac{x^3}{y}\right)$$

How to solve it

Write the known quantities and restrictions.

$$\log_b(MN) = \log_b M + \log_b N$$

Check the result against the original statement.